BADMINTON COURT MOVEMENT HEATMAP

Build Your Own Project

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1.Introduction

Badminton is played at breakneck speed, making footwork and court coverage critical to performance. Pinpointing where a player spends their time usually requires someone to sit and log video frames by hand. This project automates that workflow. Using a computer vision pipeline-combining YOLO object detection, tracking algorithms, perspective transformation, and spatial mapping-it pulls player coordinates directly from video and translates them into an intuitive court heatmap.

2.The Challenge

Trying to study how a badminton player moves by manually rewatching match footage is tedious and takes way too much time. We need a simple, automated system that uses computer vision to track a player's movement and map it out visually on a digital court.

3.Project Goals

Here is exactly what we plan to build:

- **Spot the player:** Use YOLO (a computer vision model) to automatically detect a player in the video footage.
- **Follow the action:** Keep track of that specific player as they move around during the rally.
- **Record the steps:** Pinpoint the exact coordinates of their movement.
- **Save the data:** Log all this location data neatly into a CSV spreadsheet.
- **Map it out:** Translate the camera's angled perspective into an accurate, top-down view of a standard badminton court.
- **Visualize the patterns:** Create a heatmap so you can instantly see where the player spends the most time.
- **Lay the groundwork:** Build a solid, easy-to-understand foundation for more advanced badminton analytics in the future.

4.Technologies

Python

OpenCV

NumPy

Matplotlib

Ultralytics YOLO

CSV

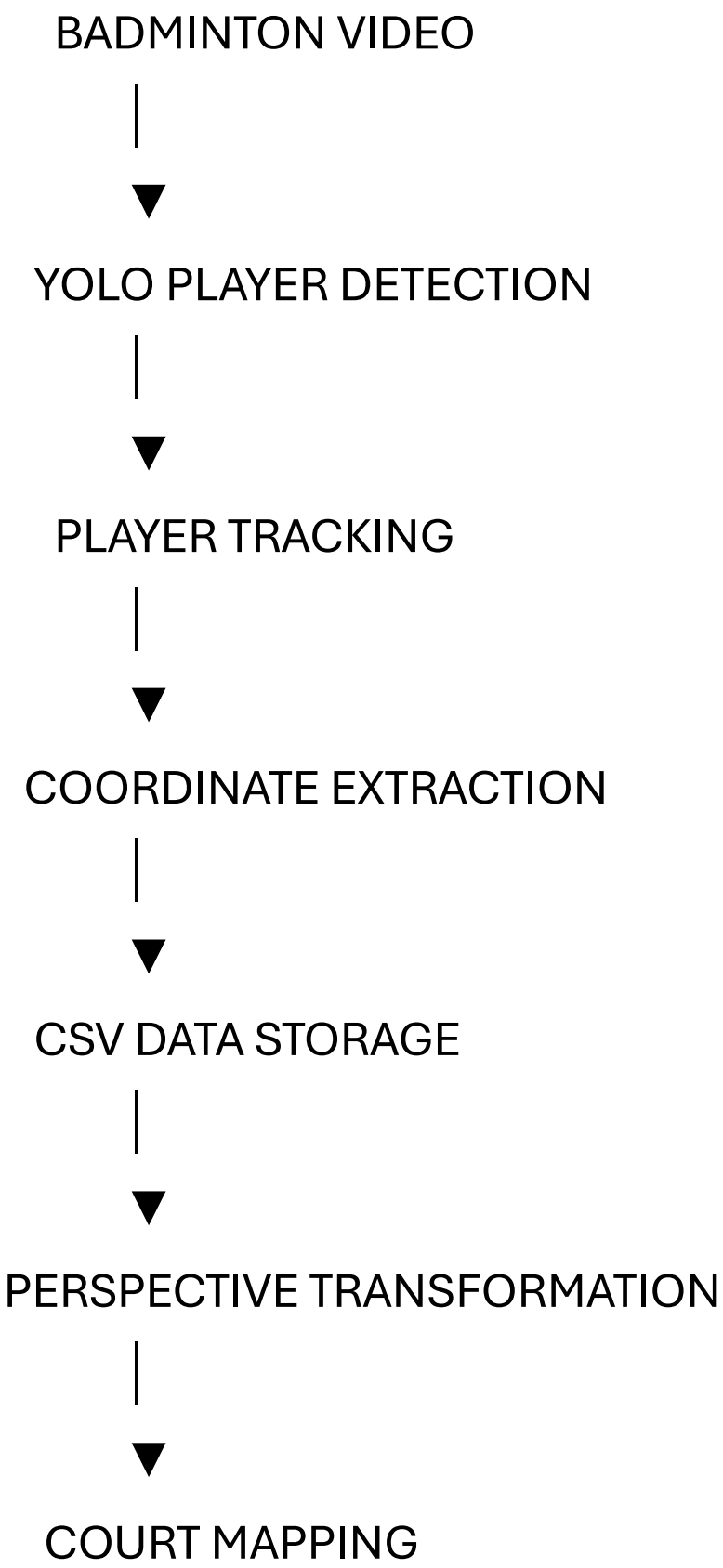
VS Code

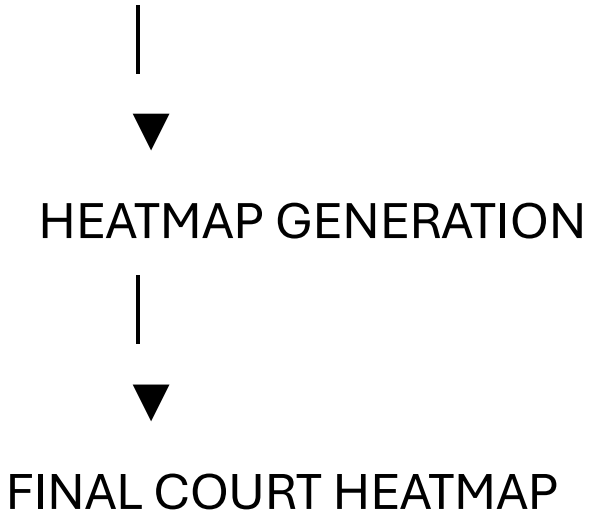
GitHub

5.Requirements

1. Accept badminton video as input.
2. Detect the player.
3. Track the player.
4. Extract coordinates.
5. Store coordinates.
6. Map coordinates to court.
7. Generate heatmap.

6.Architecture




HEATMAP GENERATION
FINAL COURT HEATMAP

7.Implementation

Module	Purpose
detect_player.py	Tests YOLO player detection
track_player.py	Tracks player and records coordinates
court_point.py	Selects court corner points
court_map.py	Performs perspective transformation and heatmap generation
heatmap.py	Generates movement visualization
save_coordinates.py	Handles coordinate storage
player_positions.csv	Stores movement data
badminton_heatmap.png	Final visualization

8.Results

Figure 1 — YOLO Player Detection

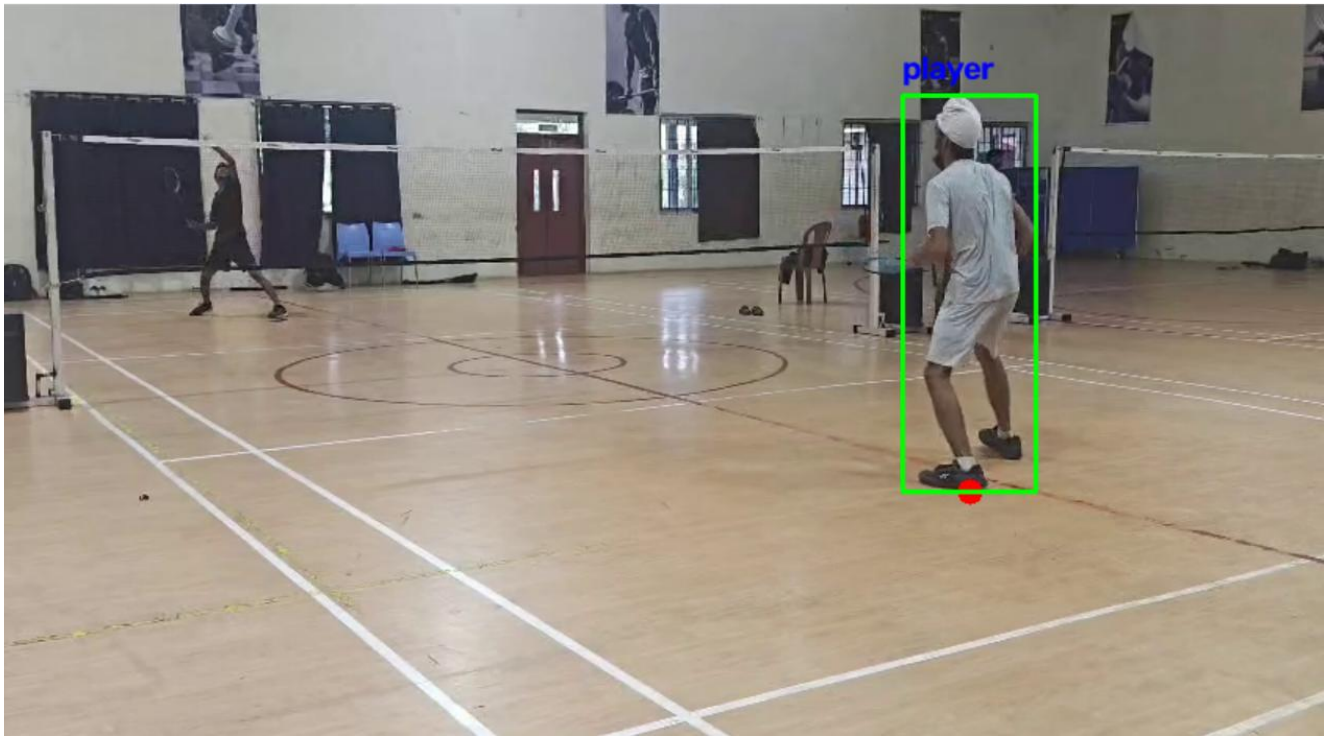
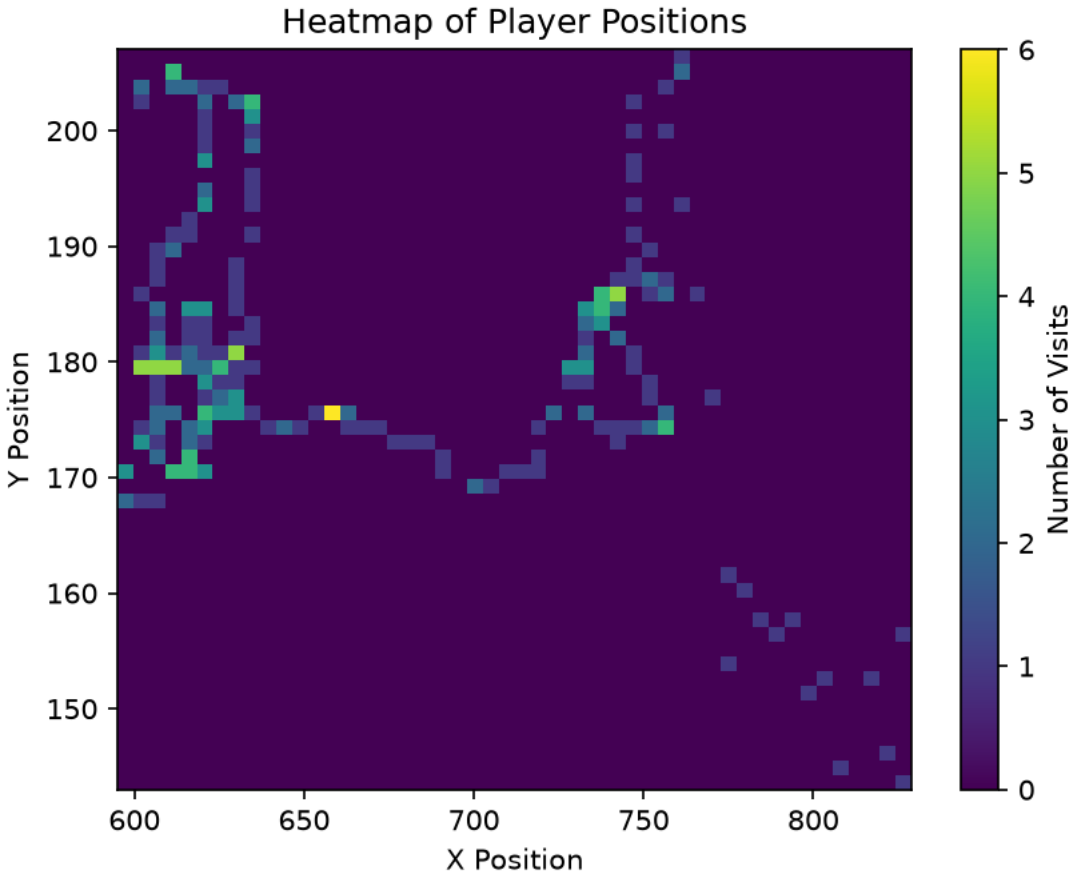
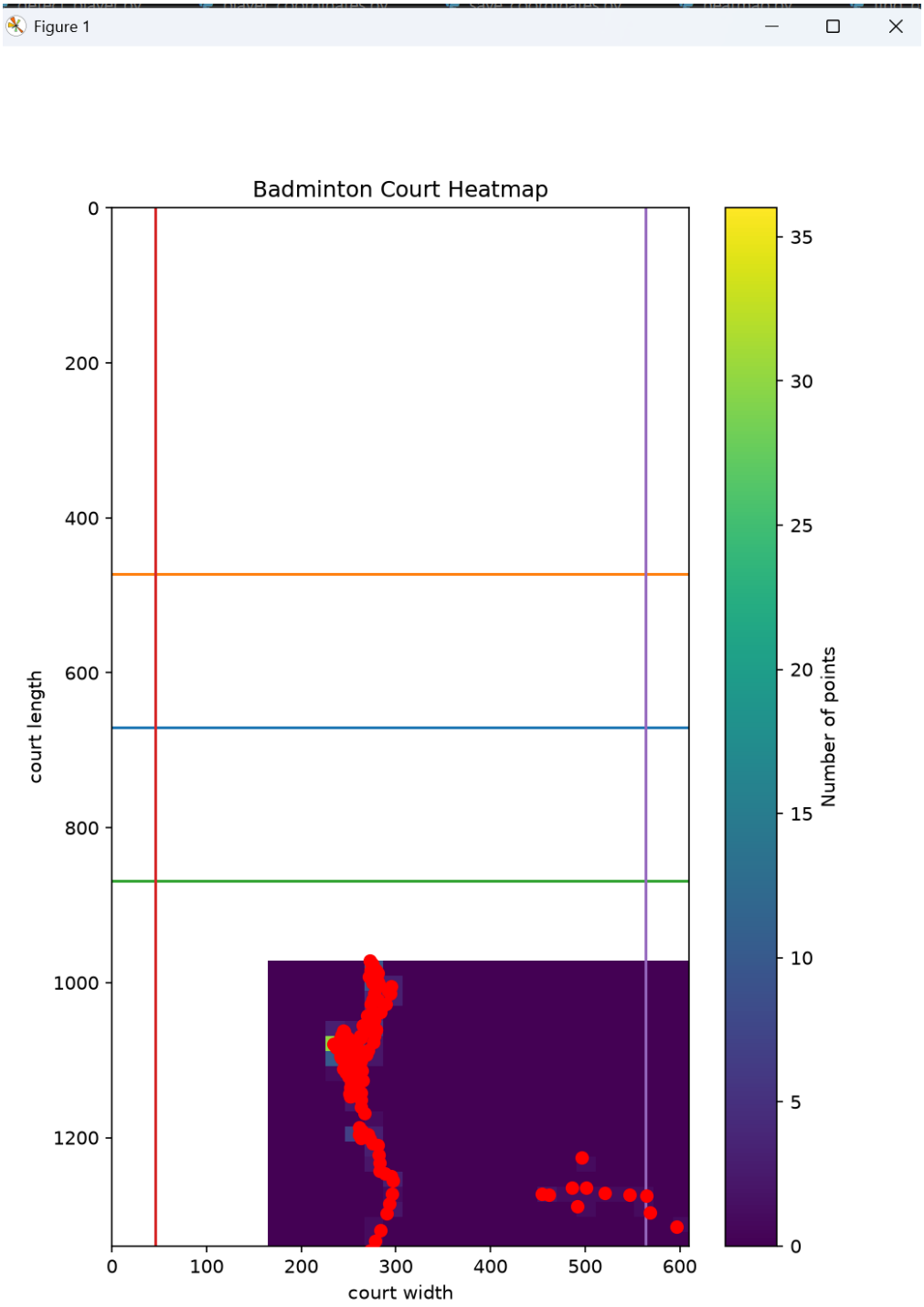


Figure 2 — Extracted Coordinates




```
PS C:\Users\ganga\OneDrive\Documents\badminton_heatmap\main> & ".\.venv\Scripts\python.exe" court_map.py
Camera X range: 595.0 to 829.0
Camera Y range: 143.0 to 207.0
Number of points: 301
X range: -15259.521 to 70324.53
Y range: -63692.07 to 16521.256
First 10 points: [(np.float32(1548.4764), np.float32(-1670.9478)), (np.float32(1729.9603), np.float32(-2043.7545)), (np.float32(1595.326), np.float32(-1757.7905)), (np.float32(1689.8068), np.float32(-1944
.7871)), (np.float32(1696.803), np.float32(-1945.8937)), (np.float32(1573.1796), np.float32(-1672.3799)), (np.float32(1573.1796), np.float32(-1672.3799)), (np.float32(1490.0238), np.float32(-1509.7454)),
(np.float32(1490.0238), np.float32(-1509.7454)), (np.float32(1515.732), np.float32(-1588.7325))]
badminton created
```

Figure 3 — Final Badminton Court Heatmap



The final heatmap represents the spatial distribution of the player's movement after transforming the

detected camera coordinates into badminton court coordinates.

9.Testing

The project was tested in multiple stages.

Test 1: Player Detection

The input badminton video was processed using the YOLO model. The player was successfully detected in the video frames.

Test 2: Player Tracking

The detected player was tracked across the video and movement coordinates were recorded.

Test 3: Coordinate Storage

The extracted coordinates were stored in player_positions.csv with separate X and Y values.

Test 4: Perspective Transformation

The camera coordinates were transformed into badminton court coordinates using OpenCV perspective transformation.

Test 5: Heatmap Generation

The transformed coordinates were visualized using a 2D histogram and movement points. The final output

was successfully generated as badminton_heatmap.png.

Result

The complete pipeline successfully converted a badminton video into a court-based player movement heatmap.

10.Challenges

Challenges Faced

During development, several challenges were encountered, including selecting appropriate court corner coordinates, tracking the correct player, handling camera coordinates and ensuring that transformed points remained within the court boundaries.

An additional challenge was ensuring that incorrect detections outside the court did not distort the final heatmap. This was addressed by restricting the tracked points to the selected court region and filtering invalid transformed coordinates.

Learnings

This project provided practical experience with Python, OpenCV, YOLO object detection, object tracking, CSV data handling, perspective transformation and data visualization.

It also demonstrated the importance of testing individual modules before combining them into a complete computer vision pipeline.

Future Enhancements

The project can be extended with:

- Multiple-player tracking
- Automatic court detection
- Real-time tracking
- Shot detection
- Player movement statistics
- Distance covered calculation
- Speed analysis
- Comparison between players
- Automatic performance reports

11.Conclusion

In the end, the Badminton Court Movement Heatmap project did exactly what we set out to achieve: it took the complex, highly technical world of computer vision and successfully applied it to the fast-paced environment of live sports. We proved that you don’t need expensive, wearable sensors to deeply understand an athlete's positioning on the court.

By simply feeding standard video footage into our system, it acts as a tireless digital observer—automatically detecting the player and keeping a lock on them through every lunge, jump, and sprint. But simply tracking a bounding box on a screen was only half the battle. The real magic happened when we extracted those raw, frame-by-frame coordinates and applied perspective transformation. This allowed us to take a regular, angled camera view and mathematically flatten it, perfectly translating the player’s real-world steps onto a clean, top-down 2D map of a standard badminton court.

The final result—the movement heatmap—is much more than just a colorful graphic or a collection of data points. It instantly tells a visual story of a player's habits, highlighting their defensive sweet

spots, where they expend the most energy, and what areas of the court they might be neglecting.

While this project is a great standalone proof of concept, its true value is what comes next. We've successfully built a solid, accessible foundation that can easily be expanded into a much more advanced analytics tool. It opens the door for smarter coaching, helping players and trainers analyze footwork, tweak game strategies, and ultimately step up their performance on the court.